Bihar Engineering University, Patna

B.Tech 7th Semester Examination 2025

Course: B.Tech | *Time: 03 Hours*
Code: 155701 | *Subject: Deep Learning* | *Full Marks: 70*

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 ***Choose the correct option / answer the following (Any seven question only):*** **[2 x 7 = 14]**

(a) Part of a biological neuron that receives signals from other neurons is called
(i) Axon (iii) Soma
(ii) Dendrite (iv) Synapse

(b) In MLP training, which algorithm computes gradients layer by layer?
(i) Forward propagation (iii) Hebbian learning
(ii) Back-propagation (iv) K-means clustering

(c) _____ helps back-propagation avoid overfitting?
(i) Increasing learning rate (iii) Large momentum only
(ii) Early stopping and regularization (iv) Removing hidden layers

(d) Optimization technique that uses a moving average of the gradient to smooth updates and navigate ravines faster is
(i) RMSProp (iii) Batch Gradient Descent
(ii) Momentum (iv) AdaGrad

(e) What is the fundamental input to the Generator network in a GAN?
(i) Real training data (iii) Random noise vector
(ii) Labeled class information (iv) Output from the Discriminator

(f) When the generator produces limited variety of samples, the problem is called
(i) Overfitting (iii) Gradient vanishing
(ii) Mode collapse (iv) Underfitting

(g) Which mechanism allows the decoder to access encoder outputs?
(i) Masked Self-Attention (iii) Cross-Attention
(ii) Positional Encoding (iv) Feed-Forward Networks

(h) Which model is famous for being "Bidirectional"
(i) GPT (iii) RNN
(ii) BERT (iv) LSTM

(i) In CNNs, what is the effect of using a stride greater than 1 during convolution?
(i) Increases output feature map size (iii) Adds more channels
(ii) Reduces output feature map size (iv) Normalizes the input

(j) GoogLeNet introduced _____ module to improve computational efficiency
(i) Residual connection (iii) Inception module
(ii) Pooling pyramid (iv) Dense block

Q.2 (a) Explain the mathematical model of an artificial neuron with commonly used activation functions. **[7]**

(b) Describe the architecture and need of a Multilayer Perceptron. Discuss the role of hidden layers in solving complex non-linear problems. **[7]**

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Q.3 (a) Explain the working principle of the back-propagation algorithm in training multilayer neural networks. [7]

(b) Design a suitable multilayer neural network architecture to solve the XOR problem. [7]

Q.4 (a) Consider training a simple linear regression model $y = wx + b$ (with no bias, $b = 0$, for simplicity, so $y = wx$) using Mean Squared Error (MSE) loss. Given the dataset as (x, y)={(1, 2), (2, 4), (3, 5)}, initial parameter $w_0 = 0$ and learning rate $\eta = 0.1$. Define the Batch MSE Cost Function $J(w)$. Calculate the gradient $\frac{dJ}{dw}$ at $w_0 = 0$. Perform one update of the weight w using Batch Gradient Descent. [7]

(b) Explain the concept of parameter norm penalties in deep learning and derive the modified cost function when L2 regularization is applied. [7]

Q.5 (a) Describe various image augmentation techniques to preserve class labels while increasing dataset diversity. [7]

(b) Explain the convolution operation in CNNs with mathematical formulation. Discuss the purpose of pooling layers in CNN architectures. [7]

Q.6 (a) Describe the architecture of LeNet in detail. Explain the contribution of LeNet to early successes in handwritten digit recognition. [7]

(b) Explain the concept of residual learning in ResNet and derive the idea behind skip connections. [7]

Q.7 (a) Analyze the vanishing gradient problem in RNNs using mathematical reasoning. [7]

(b) Describe the role of the generator and discriminator in a Generative Adversarial Networks. [7]

Q.8 (a) Explain the architecture of BERT and its use of masked language modeling and next sentence prediction during pre-training. [7]

(b) Discuss the relationship between Transformer architectures and autoencoders in representation learning. [7]

Q.9 Write short notes on ***any two*** of the following: [7 x 2 = 14]

(a) Back Propagation and Differentiation

(b) RMSProp

(c) Bidirectional RNNs

(d) Image Augmentation

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